C Programming Introduction (Detailed Guide for PDF)

What is C?

C is a general-purpose, procedural programming language used for system programming and application development.

History of C

- Developed by Dennis Ritchie in 1972
- Created at Bell Labs
- Basis for many modern languages

Why Use C?

- Fast and efficient
- Low-level memory access
- Used in operating systems and embedded systems

Features of C

- Structured programming
- Portable
- Efficient
- Rich library support

Basic C Program

```
#include <stdio.h>

int main() {
    printf("Hello, World!");
    return 0;
}
```

Variables

```
int x = 10;
float y = 5.5;
```

Data Types

- int
- float
- char
- double

Operators

- Arithmetic (+, -, *, /)
- Relational (==, !=)
- Logical (&&, ||)

Conditional Statements

```
int age = 20;

if (age > 18) {
    printf("Adult");
} else {
    printf("Minor");
}
```

Loops

For Loop

```
for (int i = 0; i < 5; i++) {
    printf("%d", i);
}
```

While Loop

```
int i = 0;
while (i < 5) {
    i++;
}
```

Functions

```
void greet() {
    printf("Hello");
}
```

Arrays

```
int numbers[3] = {1, 2, 3};
```

Pointers

```
int x = 10;
int *ptr = &x;
```

Structures

```
struct Person {
    char name[50];
    int age;
};
```

File Handling

```
FILE *fp;
fp = fopen("file.txt", "r");
```

Example Program

```
#include <stdio.h>

int main() {
    int a = 5, b = 10;
    printf("%d", a + b);
    return 0;
}
```

Advantages of C

- Fast execution
- Close to hardware
- Portable

Disadvantages of C

- No garbage collection
- Complex for beginners

Best Practices

- Use meaningful variable names
- Comment code properly
- Manage memory carefully